

Part 1: Overview & Problem Statement

Speaker: Aamir

Target time: ~1 min 15 sec

[Open on the Dashboard screen showing live stats and risk distribution.]

Good afternoon everyone. We're Team ShadowTrace, and today we're presenting an AI-powered system designed to detect and investigate money laundering networks on the blockchain.

The Problem

Every year, an estimated two to five trillion dollars is laundered globally. Criminals are increasingly using cryptocurrency and autonomous AI agents to move illicit funds through complex, multi-hop transaction chains that traditional rule-based systems simply cannot keep up with. Current compliance tools either generate massive false-positive rates, or they miss sophisticated patterns entirely.

Our Solution

ShadowTrace is a full-stack detection platform that combines graph analytics, machine learning, and rule-based heuristics into a single hybrid detection engine. Instead of choosing between explainability and accuracy, we get both.

- We analyze over 50,000 transactions across 3,000+ wallets in real time.
- Our hybrid model achieves a 98.3% F1 score -- that means high precision with minimal false positives.
- For comparison, a rule-based-only approach scores just 3.5% F1, and a pure ML model, while accurate, lacks the explainability that regulators require.
- Every detection is classified under FinCEN typologies -- layering, structuring, round-tripping, rapid relay, and fan-out collection -- so investigators know exactly what pattern they're looking at.

[Point to the model comparison chart on the Dashboard.]

As you can see on the dashboard, we display the precision, recall, and F1 scores for all three approaches side by side, along with real-time risk distribution and a live alert feed. This gives analysts an immediate, at-a-glance view of network health.

>> TRANSITION: Hand off to Zayd for the Investigation deep-dive.

Part 2: Investigation Panel

Speaker: Zayd

Target time: ~1 min 15 sec

[Switch to the Investigation Panel screen. Have a high-risk cluster selected.]

Thanks Aamir. So once our system flags a suspicious cluster, the next question is: what do investigators actually do with that information? That's where the Investigation Panel comes in.

Cluster Deep-Dive

On the left, you can see a ranked list of flagged clusters sorted by risk score. When I click into one, the panel populates with everything an analyst needs.

- A wallet risk breakdown -- a bar chart showing the top wallets in the cluster and their individual scores.
- The complete transaction history -- every internal transfer with timestamps, amounts, and the specific pattern that was flagged, whether that's layering, structuring, or rapid relay.
- An explainability panel that shows the FinCEN typology classification with a confidence percentage. This is critical -- regulators don't accept black-box outputs. They need to understand why something was flagged.

Freeze Priority & Threat Matching

We also calculate a freeze priority score from zero to one hundred, factoring in velocity, recency, volume, and severity. This tells the compliance team which clusters need immediate attention versus which can wait for the next review cycle.

- We cross-reference cluster behavior against known threat actor profiles. Right now we're matching against 51 FBI-published Lazarus Group Ethereum addresses from the Bybit hack. The system calculates a behavioral similarity score so analysts can see if a cluster is acting like a known state-sponsored actor.

SAR Generation

Finally, with one click, investigators can generate a Suspicious Activity Report narrative. We integrate with the Claude API to produce a detailed, human-readable SAR, or fall back to a structured template. This cuts report-writing time from hours to seconds.

>> TRANSITION: Hand off to Mustafa for the Timeline Analysis.

Part 3: Timeline Analysis

Speaker: Mustafa

Target time: ~1 min 15 sec

[Switch to the Timeline view. Load a suspicious cluster and prepare to hit Play.]

Thanks Zayd. So we've seen how to identify and investigate a cluster -- now let me show you how we reconstruct the story of how the money actually moved.

Animated Fund Flow Playback

The Timeline view lets us replay a cluster's entire transaction history step by step. When I hit play, you'll see each transaction animate in chronological order. The cumulative volume chart on the right builds up in real time, so you can see exactly when the bulk of the funds moved.

[Hit Play and let a few transactions animate through.]

- Each transaction is labeled with its pattern type -- so you can watch a layering sequence unfold, see structuring transactions cluster just below the ten-thousand-dollar threshold, or catch a rapid relay chain moving funds through eight-plus hops in under an hour.
- The scrubber gives full manual control. Investigators can pause at any point, step forward or backward, and examine individual transactions in the log table below.
- The transaction log highlights the currently active transaction and shows sender, receiver, amount, and pattern type -- all in real time as the playback progresses.

Why This Matters

Traditional compliance tools give you a list of flagged transactions. But money laundering is a process -- it happens over time. By visualizing the temporal sequence, analysts can see the operational tempo of the laundering network. A rapid burst of transactions in a short window tells a very different story than slow, periodic transfers spread over weeks. This temporal context is what turns a list of alerts into an actionable investigation.

>> TRANSITION: Hand off to Hassan for the Global Network view.

Part 4: Global Network Visualization

Speaker: Hassan

Target time: ~1 min 15 sec

[Switch to the Globe Network view. Slowly rotate to show city nodes and arcs.]

Thanks Mustafa. Everything we've shown so far -- the detection, investigation, and timeline -- feeds into this global network view, which maps the geographic footprint of suspicious activity.

3D Globe Visualization

What you're looking at is a real-time 3D globe with wallet nodes positioned across 24 major financial centers worldwide. Each node is color-coded by risk -- green for low, yellow for medium, orange and red for high and critical risk. The arcs connecting cities represent transaction flows between wallets in different geographic regions.

- Node size scales with transaction volume -- larger nodes mean more funds flowing through that location.
- You can click on any node to inspect wallet details: its risk score, flagged patterns, transaction count, and community membership.
- The risk slider lets analysts filter by minimum risk score, so they can zoom in on just the high-risk activity and cut through the noise.

The Big Picture

This view is designed for strategic-level analysis. When you see concentrated arcs between specific cities -- say, a heavy flow corridor between two regions -- that tells you where to focus cross-border cooperation and where to deploy additional monitoring. It turns abstract transaction data into a geographic intelligence map.

Closing

To wrap up -- ShadowTrace is a complete end-to-end platform. From real-time detection with a 98.3% F1 score, to explainable investigation panels with FinCEN-classified typologies, to temporal playback that reconstructs the laundering story, to this global view that maps the threat landscape. We built this to give financial intelligence units the tools they need to stay ahead of increasingly sophisticated laundering networks. Thank you.

[Open for Q&A.]